Heterogeneous Households and Aggregate Dynamics: Belief-Invariant Analytical Moment Recursion[†]

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Abstract

I derive analytical household policies and an exact aggregate capital law of motion in a heterogeneous-household economy with idiosyncratic productivity and return risk. Three moments—aggregate capital and two wealth-sorting covariances—are sufficient for all equilibrium allocations, and the equilibrium is invariant to household beliefs. The equilibrium jointly characterizes growth, business cycles, and distributional change along a non-stationary path. The closed-form structure enables exact inversion of U.S. data on output, capital, and hours (1950–2019) into structural shocks. The inversion reveals a sorting accelerator—TFP growth compounds return-wealth sorting, driving rising inequality—and a labor-savings tandem that accounts for several postwar recessions.

Keywords: Heterogeneous agents, closed-form aggregation, wealth inequality, business cycles, belief invariance.

JEL codes: D31, E21, E32.

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1 Introduction

Wealth inequality in the United States has risen steadily since the 1950s, yet the economy has also experienced recurring business cycles and sustained long-run growth. These three phenomena are deeply intertwined. TFP growth raises returns disproportionately for wealthy households with favorable portfolio draws, amplifying inequality through persistent return-wealth sorting. The resulting concentration, in turn, makes the macroeconomy more sensitive to aggregate shocks. Understanding this feedback requires a framework that handles growth, cycles, and distributional dynamics jointly along a non-stationary path.

This paper constructs such a framework. I show that in a heterogeneous-household economy with idiosyncratic productivity and return risk, three summary statistics—aggregate capital and two covariances measuring the alignment of wealth with productivity and portfolio returns—are sufficient for aggregate dynamics. These statistics are sufficient for all contemporaneous equilibrium allocations and prices, and aggregate capital follows an exact closed-form law of motion. I call the resulting object a *moment-recursive competitive equilibrium* (MRCE).

The paper makes three contributions. First, I derive closed-form policies for consumption, saving, and labor supply under incomplete markets with idiosyncratic productivity and return risk. These yield a unique MRCE where $\mathcal{M} = [K, \operatorname{cov}(z, a), \operatorname{cov}(e, a)]$ is sufficient for all contemporaneous equilibrium allocations and prices, and the aggregate capital law of motion K' is an exact closed-form function of $\mathcal{M}$. The saving rate is increasing and concave in wealth (Dynan et al., 2004), and the borrowing constraint is never binding. The framework nests several benchmarks: shutting down return heterogeneity yields Krusell–Smith; shutting down all heterogeneity recovers Brock and Mirman (1972). With return risk alone, the wealth distribution drops out as in Angeletos (2007); adding productivity heterogeneity restores its role through exactly two covariances.

Second, the equilibrium is invariant to household beliefs about aggregate dynamics. Under log utility, income and substitution effects cancel exactly, so policies depend only on current prices. This eliminates the fixed-point problem over beliefs, making the MRCE a belief-invariant recursive competitive equilibrium (BRCE). The BRCE accommodates non-

stationary paths without detrending—essential for studying how TFP *growth*, not just TFP levels, drives inequality. Conventional approaches either compare stationary equilibria at different TFP levels or require expectations over the entire future price sequence. The BRCE sidesteps both: any TFP path can be fed into the forward simulation to obtain valid equilibrium dynamics.

Third, the closed-form structure enables an *exact general equilibrium inversion*. Given U.S. data on output, capital, and hours (1950–2019), I recover TFP, time-varying patience, and labor productivity that exactly rationalize the data period by period. Because the recovered shocks are structural parameters in a model with distributional dynamics, they support counterfactual decompositions that isolate each shock's role in inequality growth. They also quantify how amplification depends on the current wealth distribution, extending the business-cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting.

The inversion yields several substantive findings. A "sorting accelerator" links TFP growth to rising inequality: higher TFP raises returns, which compound through near-unit-root return-wealth sorting, widening the wealth distribution via self-reinforcing feedback (Piketty, 2014). As sorting accumulates, macroeconomic fragility grows. The same 2 percent TFP shock produces a 3 percent larger impact and modestly larger cumulative losses when launched from the high-inequality 2019 state rather than 1952. Capital tax reform counterfactuals illustrate the growth-versus-inequality tradeoff: the Reagan and Bush tax cuts collectively raise output by 3.7 percent and wages by 3.1 percent, while shifting income shares toward the top. Even the bottom 10 percent gains in absolute income levels, but their share falls as the sorting mechanism disproportionately channels the growth dividend upward.

The recovered shocks reveal that TFP is the primary driver of business cycle amplitude (std ratio 0.68 vs. baseline), consistent with Krusell and Smith (1998). Labor productivity $Z_{L,t}$ contributes a +55 percent covariance share, while the nearly constant discount factor (β_t CV $= 0.75\%$, $\text{corr}(\beta_t, \log Y_t) = -0.875$) partially offsets output fluctuations (-24 percent covariance share). TFP and $Z_{L,t}$ comove strongly ($\rho(\Delta \log A, \Delta \log Z_L) = 0.122$), while $Z_{L,t}$ and β_t display a labor-savings tandem ($\rho = -0.807$). The 1973–75 episode is demand-driven (β decline) and 2007–09 is TFP-driven.

Related literature. Bewley (1986), Huggett (1993), and Aiyagari (1994) developed the foundational heterogeneous-agent framework. Krusell and Smith (1998) introduce aggregate risk and show numerically that mean wealth nearly suffices for forecasting prices; the MRCE provides an exact analytical counterpart that clarifies when and why this approximation succeeds. Computational advances include Reiter (2009), Winberry (2018), and Ahn et al. (2018). Den Haan and Rendahl (2010) establish the principle that cross-sectional averages of the individual policy's basis functions serve as aggregate state variables. The MRCE identifies the specific objects: because the saving policy $a' = \beta a(1 + \tilde{r}e + \tilde{w}Z_L\xi z) + T$ is multiplicative in (e, z), the relevant cross-sectional averages are $\mathbb{E}[ea]$ and $\mathbb{E}[za]$—i.e., the covariances $\text{cov}(e, a)$ and $\text{cov}(z, a)$, which measure return-wealth and productivity-wealth sorting. The MRCE thus provides economic content for their aggregation principle: the sufficient statistics are not generic moments but sorting measures with interpretable roles in the K' equation (17). The sequence-space Jacobian of Auclert et al. (2021) computes linearized impulse responses for general heterogeneous-agent New Keynesian economies. The MRCE delivers exact nonlinear dynamics for a specific class, and the two are complementary: SSJ applies to richer models at the cost of linearization, while the MRCE provides exact state-dependent amplification within its scope.

The analytical structure builds on exact aggregation results. Brock and Mirman (1972) solve the stochastic growth model under log utility; the MRCE nests their solution. Angeletos (2007) shows that uninsured return risk admits exact aggregation in which the wealth distribution is irrelevant for aggregate dynamics. The MRCE adds productivity heterogeneity, which brings the distribution back through covariances that serve as sufficient statistics for equilibrium allocations. Bilbiie (2020) and Ravn and Sterk (2021) develop parsimonious heterogeneous-agent models with analytical aggregate dynamics; this paper uses a continuum of agents with rich return heterogeneity. The belief-invariance result connects to Menzio and Shi (2010) on block-recursive equilibria and Gabaix (2020) on cognitive discounting. In the BRCE, invariance arises endogenously from the income-substitution offset under log utility, rather than from bounded rationality. Alvarez and Lippi (2022) develop analytical impulse responses using eigenvalue methods in menu-cost models; the analytical moment structure plays an analogous role for heterogeneous households.

The state-dependent amplification results contribute to the nonlinear business cycle literature: Storesletten et al. (2004) document countercyclical cross-sectional risk, Bloom et al. (2018) study uncertainty-driven state dependence, and Berger and Vavra (2015) analyze wealth heterogeneity in consumption dynamics. The asset pricing results relate to idiosyncratic risk and equilibrium returns (Constantinides and Duffie, 1996; Schmidt, 2016); growing evidence on heterogeneous portfolio returns (Fagereng et al., 2020; Bach et al., 2020) motivates the model's return channel.

The remainder of the paper proceeds as follows. Section 2 sets up the model environment. Section 3 derives the MRCE, establishes belief invariance, and draws out its macroeconomic and asset pricing implications. Section 4 implements the exact inversion on U.S. data and evaluates the model's quantitative predictions. Section 5 concludes. All proofs, extensions to CRRA-GHH and Epstein-Zin preferences, and supplementary results are collected in the Online Appendix.

2 Model

A continuum of infinitely lived households face idiosyncratic labor productivity and portfolio return risk and choose consumption, saving, and labor supply each period.

2.1 States and shocks

Each household's state is $x = [a, z, e]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous), and a return-scaling factor e (exogenous). All three display substantial cross-sectional variation (Kuhn et al., 2020; Storesletten et al., 2004; Fagereng et al., 2020).

The idiosyncratic processes follow stationary AR(1) dynamics in levels with unit mean:

$$z' - 1 = \rho_z (z - 1) + \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0, \sigma^2_{\varepsilon_z}), \quad \sigma^2_{\varepsilon_z} = \sigma^2_z(1 - \rho^2_z), \tag{1}$$

$$e' - 1 = \rho_e (e - 1) + \varepsilon'_e, \quad \varepsilon'_e \sim \mathcal{N}(0, \sigma^2_{\varepsilon_e}), \quad \sigma^2_{\varepsilon_e} = \sigma^2_e(1 - \rho^2_e), \tag{2}$$

where σ^2_z and σ^2_e denote the unconditional variances and $\varepsilon'_z, \varepsilon'_e$ are mutually independent innovations. Linearity in levels is essential: it ensures that the cross-sectional covariances

$\text{cov}(z, a)$ and $\text{cov}(e, a)$ serve as sufficient statistics for contemporaneous equilibrium allocations (Section 3). In the quantitative analysis, each process is discretized using the Tauchen (1986) method.

Each household's effective labor productivity per hour is $Z_L z$, where Z_L is an aggregate labor productivity parameter and z is the idiosyncratic component ($\mathbb{E}[z] = 1$). Z_L captures aggregate labor market conditions—utilization intensity, matching quality, and workforce composition—that scale effective household earnings without entering the aggregate production function. This asymmetry makes Z_L a structural shock distinct from TFP (see Section 4). At steady state, $Z_L = 1$.

The aggregate state is $X = [\Phi, A, Z_L, \beta, B, C]$: the cross-sectional distribution Φ, aggregate TFP A, aggregate labor productivity Z_L, the discount factor β, return dispersion B, and productivity dispersion C. In the quantitative application (Section 4), A, Z_L, and β vary; B and C are held at steady state. The framework accommodates time-variation in all five—e.g., widening return dispersion during financial crises or shifting patience during demand-driven episodes.

2.2 Preferences

Household preferences take a Stone-Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \tag{3}$$

where c is consumption, n is labor supply, and $\theta > 0$ governs the relative weight on leisure.

The subsistence level $g(x; X) = \kappa w(X)(1 - \tau) Z_L z$ captures work-related expenses that scale with after-tax earnings.

The effective time endowment $\kappa + \xi a$ increases in wealth: $\kappa \in (0, 1)$ is the base endowment, and $\xi > 0$ captures wealth buying usable time through time-saving services (Aguiar and Hurst, 2007). This wealth-leisure complementarity links the wealth distribution to aggregate labor supply.

The parameters (θ, κ, ξ) are calibrated to match average hours, the saving-to-labor-income ratio, and the level of wealth concentration.

2.3 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x;X) = \max_{c,a',n} \quad u(c,n,a,z;X) + \beta \mathbb{E} v(x';X') \tag{4}$$

$$\text{s.t.} \quad c + a' = w(X)(1-\tau)Z_L z n + (1 + r(X)(1-\tau)e)a \tag{5}$$

$$\Phi' = \Gamma_\Phi(X) \tag{6}$$

$$a' \geq 0, \quad n \geq 0 \tag{7}$$

The household's problem does not include the transfer T: households are unaware of its existence and optimize based solely on factor income. The actual law of motion for wealth is $a'_{\text{actual}} = a'_{\text{chosen}} + T(X)$, where T is added by the government after the household's decision (see Section 2.4). The wage $w(X)$ is per hour of labor and $r(X)$ is the aggregate return on capital. The factor e scales returns: $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average. The product $Z_L z$ determines the household's effective hourly wage; Z_L captures aggregate labor market conditions while z captures individual heterogeneity. The discount factor $\beta \in (0,1)$ is an element of the aggregate state X and may vary over time, capturing shifts in patience due to financial conditions, uncertainty, or other demand-side forces. The household faces a borrowing constraint $a' \geq 0$, which is non-binding in equilibrium (shown below).

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = A K_{eff}^{\alpha} L^{1-\alpha}, \tag{8}$$

where the inputs are *efficiency-weighted* factor aggregates:

$$K_{eff} \equiv \int e\, a\, d\Phi = K + \text{cov}(e,a), \qquad L \equiv Z_L \int z\, n(x;X)\, d\Phi, \tag{9}$$

with $K = \int a\, d\Phi$ denoting physical (wealth) capital and $\alpha \in (0,1)$. The return factor e_i scales the efficiency of household i's capital in production (e.g., reflecting portfolio quality

or entrepreneurial ability), just as z_i scales labor productivity. Factor prices satisfy:

$$r(X) = \alpha A K_{eff}^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1-\alpha) A K_{eff}^{\alpha} L^{-\alpha}, \tag{10}$$

where δ is the depreciation rate. Competitive factor markets pay $(r+\delta)$ per efficiency unit of capital and w per efficiency unit of labor. Household i supplies $e_i a_i$ efficiency units of capital and $Z_L z_i n_i$ efficiency units of labor, earning $(r+\delta)e_i a_i - \delta a_i$ and $w Z_L z_i n_i$ respectively. Aggregating: total capital income $(r+\delta)K_{eff} = \alpha Y$ and total labor income $wL = (1-\alpha)Y$, exhausting output. Because factor prices depend on K_{eff}/L, distributional forces—both labor supply shifts and return-wealth sorting—feed back into prices.

2.5 Government policy

The government levies proportional income taxes—at rate τ_K on capital income and τ_L on labor income—and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot \big[\tau_K (r+\delta) K_{eff} + \tau_L w L\big], \tag{11}$$

When $\tau_K = \tau_L = \tau$, this reduces to $T = \eta \tau Y$ (since $(r+\delta)K_{eff} + wL = Y$). After-tax prices are $\tilde{r} = r(1-\tau_K)$ and $\tilde{w} = w(1-\tau_L)$.[1] Remaining revenue finances government consumption (not in utility or production). The baseline sets $\tau_K = \tau_L = 0.30$; Section 4.8 and Online Appendix E.10 consider time-varying rates.

The transfer plays a dual role. First, it is a fiscal policy channel: tax revenue is partially redistributed, with η controlling the degree of redistribution. Higher η compresses the wealth distribution and thins the Pareto tail. Second, it provides the additive intercept in the Kesten process governing wealth dynamics, which is necessary for a stationary distribution with a Pareto tail to exist (Kesten, 1973). Without $T > 0$, the purely multiplicative system $a' = \beta R_i a$ collapses to the degenerate distribution at zero and admits no non-degenerate

[1] The transfer formula (11) taxes gross capital income $(r+\delta)K_{eff}$, while the household's after-tax return $\tilde{r} = r(1-\tau_K)$ is defined on the net return r. This simplification keeps the household problem tractable; the quantitative difference is $\tau_K \delta \approx 1.5$ percentage points at baseline values.

stationary distribution.

Timing. The transfer arrives after the household's consumption-saving decision: households solve their problem without knowledge of T, and the government adds T to wealth at the end of the period. This timing is essential for preserving the analytical tractability. If households anticipated T, they would incorporate its present value into effective wealth, breaking the $v = A \log a + B$ value function conjecture and the multiplicative-additive separation $a' = \beta R_i a + T$ that enables closed-form aggregation.

2.6 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_Φ.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. *Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ.*

The difficulty is that Φ is infinite-dimensional. The next section shows that optimal policies depend on Φ only through three moments—aggregate capital and two covariances—that serve as sufficient statistics for all contemporaneous equilibrium allocations.

3 Moment-Recursive Competitive Equilibrium

3.1 Analytical household policies

The household's optimal policies have explicit solutions that reduce the infinite-dimensional distribution to a finite moment vector.

Proposition 1 (Analytical household policies and wealth dynamics)**.**
The optimal consumption and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1+r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)Z_L z \tag{12}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)Z_L z},\ 0\right\} \tag{13}$$

The household's chosen saving is $\hat{a}'(x;X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right)$*. The actual law of motion for wealth, after the unanticipated government transfer, is:*

$$a'(x;X) = \hat{a}'(x;X) + T(X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) + T(X). \tag{14}$$

Proof. The budget constraint can be rewritten as:

$$(1+\theta)(c - \kappa Z_L z w(X)(1-\tau)) + a' = \left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) a. \tag{15}$$

Under log utility, the standard result of Brock and Mirman (1972) yields $a' = \beta a R_i$. The government then adds the unanticipated transfer: $a'_{\text{actual}} = \beta a R_i + T(X)$. ■

Consumption is linear in wealth with source-dependent MPCs. The saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component (enabling aggregation) and an additive transfer (ensuring stationarity).

3.2 The saving rate

The saving rate has a closed-form characterization.

Proposition 2 (Analytical saving rate)**.**
The saving rate $s(x;X) = \frac{a'(x;X)-a}{y(x;X)}$*, where* $y(x;X) = w(X)(1-\tau)Z_L z n(x;X) + r(X)(1-\tau)ea + T(X)$ *is after-tax income plus transfer, satisfies:*

$$s(x;X) = \frac{a[\beta(1+\tilde{r}\,e + \tilde{w}\,Z_L z\xi) - 1] + T}{\tilde{w}\,Z_L z\,\kappa\theta + a\left[\frac{\tilde{w}\,Z_L z\,\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+\tilde{r}\,e)}{1+\theta} + \tilde{r}\,e\right] + T}, \tag{16}$$

where $\tilde{r} = r(X)(1-\tau)$ *and* $\tilde{w} = w(X)(1-\tau)$*, with the following properties:*

(i) $\lim_{a\to\infty} s(x;X) = \dfrac{\beta(1+\tilde{r}\,e+\tilde{w}\,Z_L z\xi)-1}{\frac{\tilde{w}\,Z_L z\,\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+\tilde{r}\,e)}{1+\theta} + \tilde{r}\,e}$

(ii) $\lim_{a\to 0} s(x;X) > 0$ *due to the transfer* $T(X) > 0$

(iii) $\frac{\partial s}{\partial a} > 0$ *and* $\frac{\partial^2 s}{\partial a^2} < 0$.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive at zero wealth (due to T), increasing, and concave—the pattern documented by Dynan et al. (2004). Concavity generates mean reversion in relative wealth: as wealth rises, the subsistence floor absorbs a diminishing share and the saving rate converges to a constant.

3.3 Non-binding borrowing constraint

An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint)**.**
The borrowing limit $a' \geq 0$ *is always non-binding.*

Proof. Given $a'(x;X) = \beta a((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi) + T(X) \geq T(X) > 0$, the result is immediate. ■

Non-binding constraints keep policies globally linear, enabling analytical aggregation. This is a scope condition: the MRCE applies when transfers are large enough relative to borrowing needs that the constraint never binds. Where binding constraints create kinks, the analytical aggregation breaks down. The condition is broadly consistent with the U.S. economy given Social Security, Medicare, and other social insurance.

3.4 Existence and characterization

Definition 2 (Moment-recursive competitive equilibrium (MRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ *is a moment-recursive competitive equilibrium if:*

1. *g_c, g_a, g_n, v solve the household's problem given Γ_M.*

2. *K, L satisfy the production sector's optimality conditions.*

3. *Factor markets clear.*

4. *Γ_M satisfies $\Gamma_M(\mathcal{M}(\Phi), A, Z_L, \beta) = \mathcal{M}(\Phi')$, where $\mathcal{M}$ extracts the sufficient moments from Φ, and Φ' is the distribution of individual states induced by g_a and the exogenous processes.*

The key departure from the RCE is that Γ_M maps a *finite* moment vector to contemporaneous equilibrium allocations and prices. The sufficient statistics are aggregate capital K and two covariances: $\text{cov}(z, a)$ measures the alignment between earnings ability and wealth; $\text{cov}(e, a)$ measures the alignment between portfolio returns and wealth.

Proposition 3 (Unique existence of the MRCE)**.**
The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a), cov(e, a)]$. Define after-tax prices $\tilde{r} = r(\mathcal{M}, A)(1 - \tau)$, $\tilde{w} = w(\mathcal{M}, A)(1 - \tau)$, and the transfer $T = \eta \cdot Y \cdot \tau$. Then:

(i) Contemporaneous sufficiency. Given $\mathcal{M}$ and the aggregate state (A, Z_L, β), all equilibrium allocations—individual consumption $c(x; X)$, saving $a'(x; X)$, labor supply $n(x; X)$, and aggregate prices r, w, L—are fully determined without reference to the distribution Φ beyond its moments $\mathcal{M}$.

(ii) Aggregate capital law of motion. K' has an exact closed-form expression:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi\, cov(z, a) + \tilde{r}\, cov(e, a)\right] + T \tag{17}$$

(iii) Factor prices and aggregate labor. Defining $K_{eff} = K + cov(e, a)$:

$$r(\mathcal{M}, A) = \alpha A\left(\frac{L}{K_{eff}}\right)^{1-\alpha} - \delta \tag{18}$$

$$w(\mathcal{M}, A) = (1 - \alpha)A\left(\frac{K_{eff}}{L}\right)^{\alpha} \tag{19}$$

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L\kappa + \frac{Z_L\xi(1 + \theta\beta)(K + cov(z, a))}{1 + \theta} - \frac{\theta(1 - \beta)}{(1 + \theta)\tilde{w}}[K(1 + \tilde{r}) + cov(e, a)\tilde{r}] \tag{20}$$

Proof. See Appendix A.3 of the Online Appendix. ■

The K' equation reveals three distinct channels. The first term $(1 + \tilde{r} + \tilde{w} Z_L \xi)K$ is the representative-agent channel. The second, $\tilde{r}\,\mathrm{cov}(e, a)$, is the *return-sorting channel*: high-return households are disproportionately wealthy, raising aggregate saving above what average returns and average wealth would predict. The third, $\tilde{w} Z_L \xi\,\mathrm{cov}(z, a)$, is the *productivity-sorting channel*: productive-wealthy households generate additional saving through wealth-leisure complementarity. These terms make dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional dynamics and the closure problem. While $\mathcal{M}$ is sufficient for all contemporaneous equilibrium allocations and K' is exactly determined, the forward evolution of $\mathrm{cov}(z', a')$ and $\mathrm{cov}(e', a')$ generally depends on higher-order moments of the joint distribution Φ. Specifically, from $a' = \beta a\big((1 + \tilde{r}e) + \tilde{w} Z_L z \xi\big) + T$ and $z' = \rho_z z + \varepsilon_z$:

$$\mathrm{cov}(z', a') = \rho_z \beta \big[(1 + \tilde{r} + \tilde{w} Z_L \xi)\,\mathrm{cov}(z, a) + \tilde{w} Z_L \xi\,\mathbb{E}[\hat{z}^2 a] + \tilde{r}\,\mathbb{E}[\hat{z}\hat{e}a]\big], \tag{21}$$

where $\hat{z} = z - 1$ and $\hat{e} = e - 1$. The terms $\mathbb{E}[\hat{z}^2 a]$ and $\mathbb{E}[\hat{z}\hat{e}a]$ involve three-way cross-moments that are not functions of $\mathcal{M}$ alone. Under joint normality of $(\hat{z}, \hat{e}, a)$, these reduce to $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$ and $\mathbb{E}[\hat{z}\hat{e}a] = 0$ (by the Isserlis/Wick theorem, using independence of z and e), and the moment dynamics close. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality.

Therefore, tracking the full distribution via agent-based simulation is necessary for the forward evolution of distributional moments, while the exact household policies, the K' equation (17), and all contemporaneous allocations are preserved analytically. The quantitative analysis in Section 4 employs this approach: exact policies govern each agent's behavior, and the distributional moments are computed from the simulated cross-section.

3.5 Ergodicity

Corollary 2 (Kesten stationarity)**.**
The individual law of motion $a' = \beta a\big((1 + \tilde{r}e) + \tilde{w} Z_L z \xi\big) + T$ *is a Kesten process with*

multiplicative coefficient $R_i = \beta\big((1+\tilde{r}e) + \tilde{w}Z_L z\xi\big)$ *and additive intercept* $T > 0$. *When* $\mathbb{E}[\log R_i] < 0$, *the wealth distribution admits a unique stationary solution with Pareto tails (Kesten, 1973). The additive transfer* $T > 0$ *is essential: without it, the purely multiplicative system admits only the degenerate distribution at zero.*

Proof. See Appendix A.4 of the Online Appendix. ■

The near-unit-root property of the dominant eigenvalue (empirically ≈ 1.01 in the calibrated model) generates slow mean reversion of the wealth distribution and amplifies shock propagation, as shown in Section 4.

3.6 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue)**.**
Suppose return heterogeneity is muted: $e = 1$. *Then, the MRCE uniquely exists with* $\mathcal{M}(\Phi) = [K, cov(z,a)]$, *and the aggregate capital law of motion is:*

$$K' = \beta\left[(1+\tilde{r}+\tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi\, cov(z,a)\right] + T \tag{22}$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \mathrm{cov}(z,a)]$—the analytical counterpart of Krusell and Smith (1998). Capital alone suffices if and only if $\mathrm{cov}(z,a) = 0$. Approximate aggregation holds when $\mathrm{cov}(z,a)$ is small relative to K—i.e., when productivity shocks are weakly persistent or wealth mobility is high. The MRCE thus provides exact conditions under which the Krusell and Smith (1998) finding holds as a theorem. Section 4 confirms: the representative-agent channel accounts for 99.6 percent of mean investment. Further discussion appears in Appendix B.3.

Corollary 4 (Representative-agent analogue)**.**
Suppose all heterogeneity is muted: $e = 1$ *and* $z = 1$. *Then, the MRCE uniquely exists with*

$\mathcal{M}(\Phi) = K$, *and the law of motion is:*

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w} Z_L \xi)K\right] + T, \tag{23}$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A, Z_L, \beta) = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w} Z_L \xi)K + \kappa \tilde{w} Z_L. \tag{24}$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mirman, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

3.7 Belief invariance

A distinctive feature of the MRCE is invariance to household beliefs about aggregate dynamics, eliminating the fixed-point problem that dominates standard heterogeneous-agent computation.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ *is a belief-invariant recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.*

2. K, L *satisfy the production sector's optimality conditions.*

3. *Factor markets clear.*

4. Γ_Φ *satisfies* $\Gamma_\Phi(X) = \Phi'$, *where* Φ' *is the future distribution of individual states constructed by applying the policy function* g_a *and the exogenous idiosyncratic process to* Φ.

Theorem 1 (Belief invariance).

In the BRCE, the law of motion Γ_Φ is unaffected by households' expectations about future aggregate states.

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x;X) = \beta a((1+r(X)(1-\tau)e)+w(X)(1-\tau)Z_L z\xi)+T(X)$ depends only on current prices and Z_L—not on forecasts. Under log utility, income and substitution effects of future return changes exactly offset, so all policies are belief-invariant and the aggregate law of motion inherits this property.

Proposition 4 (Unique existence of the BRCE).

The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

Because the MRCE and BRCE coincide, the equilibrium is computed by forward simulation from any initial distribution Φ_0 and aggregate sequences $\{A_t, Z_{L,t}, \beta_t\}$—no fixed-point iteration, no assumption on expectations, no stationarity requirement. Each period, the exact household policies determine individual decisions, aggregate moments are computed from the cross-section, and the distribution evolves forward. The same procedure applies whether aggregates follow stationary processes, deterministic trends, or both. This also enables the exact inversion: structural shocks can be recovered from data without taking a stand on expectation formation.

Remark 1 (Saving share under time-varying β).

Since β is an element of the aggregate state X, it may vary over time. With time-varying β, the value function takes the form $v(a,z,e;X) = \zeta \log a + B(z,e,X)$, where ζ satisfies $\zeta = (1+\theta) + \beta\,\mathbb{E}[\zeta']$. Under belief invariance, households treat the current β as permanent, yielding $\zeta = (1+\theta)/(1-\beta)$ and saving share $s = \beta$. All propositions above hold with the current β from X: the saving policy is $a'(x;X) = \beta\, a\big((1+\tilde{r}e)+\tilde{w}Z_L\xi z\big)+T$, and the aggregate capital law of motion is $K' = \beta\big[(1+\tilde{r}+\tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi\, cov(z,a) + \tilde{r}\, cov(e,a)\big] + T$, with no additional state variables or expectation formation required. See Online Appendix A.8 for the ζ recursion derivation.

3.8 Macroeconomic implications

State-dependent aggregate dynamics. The capital law of motion (17) decomposes into three channels: the representative-agent channel $(1+\tilde{r}+\tilde{w}Z_L\xi)K$, the return-sorting channel $\tilde{r}\,\mathrm{cov}(e,a)$, and the productivity-sorting channel $\tilde{w}Z_L\xi\,\mathrm{cov}(z,a)$. These distributional terms make aggregate dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional channels. The aggregate consumption function decomposes as:

$$C(\mathcal{M},A,Z_L,\beta)=\underbrace{\frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}Z_L\xi)K+\kappa\tilde{w}Z_L}_{\text{Representative-agent level}}+\underbrace{\frac{1-\beta}{1+\theta}\left(\tilde{r}\,\mathrm{cov}(e,a)+\tilde{w}Z_L\xi\,\mathrm{cov}(z,a)\right)}_{\text{Distributional channel}} \tag{25}$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M},A,Z_L,\beta)=\kappa+\underbrace{\frac{Z_L\xi(1+\theta\beta)(K+\mathrm{cov}(z,a))}{1+\theta}}_{\text{Leisure enhancement}}-\underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r})+\tilde{r}\ \overbrace{\mathrm{cov}(e,a)}^{\text{Dist. channel}}\ ]}{(1+\theta)\tilde{w}Z_L}}_{\text{Wealth effect}} \tag{26}$$

The wealth-effect term is amplified by $\mathrm{cov}(e,a)$: wealthy households earning high returns consume more leisure, reducing labor supply. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response as the households with the largest wealth gains withdraw the most labor.

State-dependent inequality dynamics. The K' equation (17) generates a "wealth spiral": productivity shocks raise capital, which strengthens return-wealth sorting through $\mathrm{cov}(e,a)$, which raises future capital. Discounting ($\beta<1$) and the transfer T counterbalance. The persistence of the sorting mechanism—governed by the near-unit-root dynamics of the wealth distribution—controls the speed of concentration (Appendix B.4).

3.9 Asset pricing implications

Individual return heterogeneity, traditionally viewed as diversifiable, carries a positive risk premium because wealth sorting gives it aggregate consequences (Constantinides and Duffie,

1996; Storesletten et al., 2004). Households with persistently high returns accumulate disproportionate wealth, shifting labor supply and equilibrium returns:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \tag{27}$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

4 Quantitative Analysis

The MRCE's analytical structure enables an *exact general equilibrium inversion*: three aggregate equations in three unknowns, solved period by period. The recovered shocks match observed GDP, capital, and hours *exactly* by construction. The theory yields five predictions that discipline the recovered shocks:

P1. *Approximate aggregation.* The representative-agent channel dominates investment dynamics; distributional covariances contribute only at the margin (Section 3.8).

P2. *State-dependent amplification.* The same aggregate shock produces larger macroeconomic responses when wealth inequality—measured by $\text{cov}(e, a)$—is high (Section 3.8).

P3. *Return risk premium.* Higher return-wealth sorting $\text{cov}(e, a)$ lowers the equilibrium interest rate, creating an endogenous channel through which idiosyncratic return dispersion affects aggregate prices (Section 3.9).

P4. *Countercyclical patience.* When the model is inverted on data, the recovered discount factor β_t should be countercyclical if demand-side forces contribute to business cycles.

P5. *Near-unit-root persistence.* The dominant eigenvalue governing distributional dynamics is close to one, implying that wealth sorting is extremely persistent and acts as an internal propagation mechanism (Section 3.8).

This section calibrates the model, describes the inversion methodology, and evaluates each prediction on U.S. data from 1950 to 2019.

4.1 Calibration and inversion methodology

Calibration. Table 1 reports parameter values at annual frequency. The capital share $\alpha = 0.33$ follows Cooley and Prescott (1995). The depreciation rate δ_t is time-varying, computed from BEA Fixed Assets tables (FRED M1PTOTL1ES000 / K1PTOTL1ES000) as the ratio of current-cost depreciation to the net stock of private fixed assets. It rises from 0.037 in 1950 to 0.058 in 2019 (mean 0.050), reflecting the secular shift from long-lived structures to shorter-lived equipment and intellectual property.

The discount factor $\beta = 0.96$ is standard; the leisure weight $\theta = 4$ is calibrated so that the sequential inversion recovers near-constant $(\beta_t, Z_{L,t})$ when applied to Krusell–Smith (1998) simulated data (see validation below); $\kappa = 0.27$ is set below the minimum observed labor supply to ensure interior solutions; $\xi = 0.0213$ targets a saving-to-labor-income ratio of 0.33. The baseline tax rate $\tau = 0.30$ represents the effective U.S. average; Section 4.8 and Online Appendix E.10 consider time-varying rates from Mendoza et al. (1994). The transfer share $\eta = 0.0133$ targets the level of wealth concentration (Gini ≈ 0.80). Idiosyncratic productivity follows Storesletten et al. (2004): $\rho_z = 0.95$, $\sigma_z = 0.12$. Return parameters follow Fagereng et al. (2020): $\rho_e = 0.999$, $\sigma_e = 0.01$.

The model incorporates countercyclical dispersion. Following Storesletten et al. (2004), the innovation standard deviation $\sigma_{z,t}$ amplifies by a factor of 1.75 when $Z_{L,t}$ declines. Following Bloom et al. (2018), $\sigma_{e,t}$ amplifies by a factor of 4.1 when output declines. These multipliers affect the distributional dynamics—and hence $K_{eff,t}$ and L_t—through the agent-based simulation, feeding back into the recovered $(A_t, Z_{L,t}, \beta_t)$ via the within-period equilibrium.

Exact general equilibrium mapping. The MRCE's aggregate equations can be inverted period by period, extending the business cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting with structural (rather than reduced-form) shocks. The three

Table 1: Calibrated parameters

Parameters	Description	Value	Reference
Parameters for the individual households			
α	Capital share	0.3300	
β	Discount factor	0.9600	
δ_t	Depreciation rate (time-varying)	0.037–0.058	BEA Fixed Assets
θ	Labor disutility	4.0000	KS98 validation
κ	Common time endowment	0.2700	$\kappa < \min(L_{\text{data}})$
ξ	Wealth–leisure complementarity	0.0213	Saving/Labor Income ≈ 0.33
ρ_z	Idiosyncratic prod. persistence	0.9500	Storesletten et al. (2004)
ρ_e	Idiosyncratic return persistence	0.9990	Fagereng et al. (2020)
σ_z	Idiosyncratic prod. volatility	0.1200	Storesletten et al. (2004)
σ_e	Idiosyncratic return volatility	0.0100	Fagereng et al. (2020)
Government policy parameters			
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Parameters for the business cycle			
ρ_A	Aggregate TFP persistence	0.9500	Cooley and Prescott (1995)
ρ_{σ_e}	Uncertainty shock persistence	0.7800	Bloom et al. (2018)
σ_A	Aggregate TFP volatility	0.0070	Cooley and Prescott (1995)
σ_{σ_e}	Uncertainty shock volatility	0.3600	Bloom et al. (2018)

Notes: Calibration is based on annual frequency. The capital share $\alpha = 0.33$ follows Cooley and Prescott (1995); the depreciation rate δ_t is time-varying from BEA private fixed assets data (see main text). The discount factor $\beta = 0.96$ is the standard annual value. Labor disutility $\theta = 4$ is calibrated via KS98 validation (see main text). The base time endowment $\kappa = 0.27$ is set below the minimum observed labor supply to ensure interior solutions. The wealth-leisure parameter $\xi = 0.0213$ targets a saving-to-labor-income ratio of approximately 0.33. The proportional tax rate $\tau = 0.30$ represents the effective U.S. tax rate. The transfer share $\eta = 0.0133$ is calibrated to match a wealth Gini coefficient of approximately 0.80. Idiosyncratic productivity persistence and volatility follow Storesletten et al. (2004); idiosyncratic return parameters follow Fagereng et al. (2020).

key aggregate equations are:

$$Y_t = A_t K_{eff,t}^{\alpha} L_t^{1-\alpha} \tag{28}$$

$$L_t = \kappa + Z_{L,t}\,\tilde{\xi}_t \frac{1+\theta\beta_t}{1+\theta}(K_t + \text{cov}_t(z,a)) - \frac{\theta(1-\beta_t)}{(1+\theta)\,\tilde{w}_t Z_{L,t}}\big(K_t(1+\tilde{r}_t) + \tilde{r}_t\,\text{cov}_t(e,a)\big) \tag{29}$$

$$K_{t+1} = \beta_t\left[(1+\tilde{r}_t + \tilde{w}_t Z_{L,t}\tilde{\xi}_t)K_t + \tilde{r}_t\,\text{cov}_t(e,a) + \tilde{w}_t Z_{L,t}\tilde{\xi}_t\,\text{cov}_t(z,a)\right] + T_t \tag{30}$$

where $K_{eff,t} = K_t + \text{cov}_t(e, a)$ is effective capital, $\tilde{\xi}_t = \xi$ is the wealth-leisure complementarity parameter (held constant), and $Z_{L,t}$ is the time-varying aggregate labor productivity introduced in Section 2; at steady state $Z_L = 1$. The production function uses K_{eff}—not K—because capital efficiency varies across households (Section 2). Note that equation (29) is implicit in L: the after-tax prices $\tilde{r}$ and $\tilde{w}$ depend on L through the marginal products. The equation admits a unique solution by monotonicity of the labour supply in the wage.

Given data on $\{Y_t, K_t, L_t\}$ and initial distributional conditions, the inversion recovers $(A_t, \beta_t, Z_{L,t})$ in four sequential steps. A preliminary two-series inversion provides the covariance path $\text{cov}_t(e, a)$ needed to compute effective capital $K_{eff,t} = K_t + \text{cov}_t(e, a)$. For each period t: (i) A_t is identified from the production function $Y_t = A_t K_{eff,t}^{\alpha} L_t^{1-\alpha}$; (ii) prices r_t and w_t follow from marginal products; (iii) β_t is recovered from the capital accumulation equation (30); (iv) $Z_{L,t}$ is solved analytically from the labour supply equation (29). Because all three data series enter the production function, the original (Y_t, K_t, L_t) are recovered exactly upon re-aggregation. The sequential structure avoids the near-collinear joint identification that arises when $Z_{L,t}$ and β_t are solved simultaneously.

The recovered shocks have distinct structural interpretations. A_t captures neutral total factor productivity. β_t captures time-varying patience—or equivalently, demand-side forces operating through the saving margin. $Z_{L,t}$ is a model primitive (Section 2), not a reduced-form residual. Three features distinguish it from the labor wedge of Chari et al. (2007). First, it has a direct structural interpretation: time variation maps to shifts in labor force composition, matching quality, or utilization intensity. Second, $Z_{L,t}$ scales household decisions but not the production function, making it separately identified from TFP. Third, it has distributional consequences: $Z_{L,t}$ enters the saving policy and equilibrium prices, so shifts in labor productivity alter the dynamics of wealth inequality.

Krusell–Smith validation. The inversion is validated against Krusell and Smith (1998) simulated data, where the only aggregate shock is TFP ($A \in \{0.99, 1.01\}$) while β and Z_L are constant by construction. I simulate the KS98 economy for 1,100 periods and apply the sequential inversion to the simulated (Y_t, K_t, L_t). Figure 1 and Table 2 summarize the results. The inversion correctly recovers A_t (correlation with true $A = 1.000$; panel (a)

shows exact 45-degree alignment), while β_t and $Z_{L,t}$ remain near-constant (panels (b)–(c); $\mathrm{CV}(\beta) < 0.12\%$, $\sigma(\Delta \log Z_L) < 0.17\%$). Two extensions confirm robustness: the KS98 economy with endogenous labour supply and with Frisch labour supply and convex adjustment costs (Table 2). In all three variants, the inversion attributes virtually all aggregate dynamics to TFP, confirming that it does not spuriously generate preference or labour shocks.

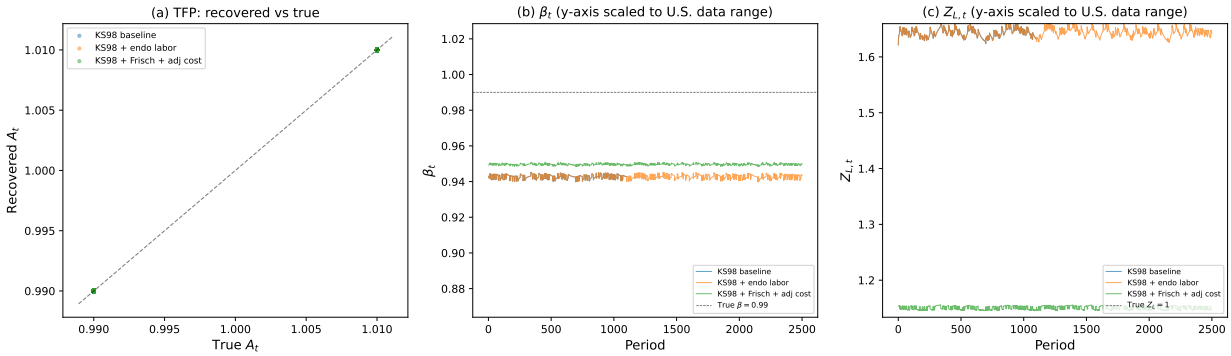

Figure 1: Krusell–Smith validation: recovered shocks from simulated data

Notes: Sequential inversion applied to simulated (Y_t, K_t, L_t) from three KS98 variants. (a) Recovered A_t vs. true A_t: exact 45-degree alignment. (b) Recovered β_t: the y-axis is scaled to the U.S. data range—the KS98 series appear as flat lines, confirming near-constant preferences. (c) Recovered $Z_{L,t}$: y-axis likewise scaled to the U.S. data range. The KS98 variation is invisible at this scale, underscoring that the inversion does not spuriously generate labour shocks.

Table 2: Krusell–Smith validation: inversion accuracy

	$\mathrm{mean}\lvert A^{\mathrm{inv}} - A^{\mathrm{true}}\rvert / A$	$\mathrm{CV}(\beta)$	$\sigma(\Delta \log Z_L)$	$\mathrm{mean}\lvert Y^{\mathrm{inv}} - Y^{\mathrm{true}}\rvert / Y$
KS98 baseline	7.5×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + endo labor	7.3×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + Frisch + adj cost	2.5×10^{-5}	0.05%	0.18%	$< 10^{-15}$
U.S. data (1950–2019)	—	1.78%	3.42%	exact

Notes: The sequential inversion is applied to simulated data from three KS98 variants. In all variants, β and Z_L are constant in the true DGP; only A varies. $\mathrm{CV}(\beta) = \sigma(\beta)/\bar{\beta}$ measures preference-shock variation. $\sigma(\Delta \log Z_L)$ measures labour-shock variation. The final row shows U.S. data results for comparison: β fluctuates 15× more and Z_L fluctuates 20× more than in KS98 simulations.

Applying the same inversion to U.S. data reveals economically significant departures from the KS98 benchmark (Table 2, last row). The discount factor β_t fluctuates 15× more than in KS98 simulations (CV = 1.78% vs. 0.12%), while $Z_{L,t}$ fluctuates 20× more ($\sigma(\Delta \log Z_L) = 3.42\%$ vs. 0.17%). The labour wedge $Z_{L,t}$ is the dominant departure: U.S.

capital accumulation is well-approximated by a single TFP shock with near-constant preferences, but labour market fluctuations require an additional structural shock that the KS98 framework cannot generate.

The inversion uses three U.S. data series over 1950–2019: real per-capita GDP (FRED GDPCA), the Penn World Tables per-capita capital stock (RKNANPUSA666NRUG), and aggregate hours (FRED HOANBS, annualized). GDP and capital are divided by a common scale factor (per-capita GDP in 1950 divided by Y_{ss}); hours are normalized so that $L_0 = L_{\text{ss}}$. The time-varying capital share α_t and depreciation rate δ_t are taken from Penn World Tables and BEA Fixed Assets data, respectively; the analytical expressions carry over with α replaced by α_t period by period, since each period's equilibrium is contemporaneous. The data K/Y ratio starts at 4.34 in 1950 and declines to 3.65 by 2019. The initial $A_0 = 0.802$ adjusts to reconcile the data capital-output ratio with the production function using $K_{eff,0} = K_0 + \text{cov}_0(e, a)$, where the initial covariance is from a 500-agent burn-in at steady-state prices. Results are robust to alternative initializations.

4.2 Data-anchored results

Figure 2 displays the three recovered structural series.

Aggregate TFP. TFP grows from 0.802 in 1950 to 1.78 in 2019 (1.16 percent per year), tracing the postwar expansion, 1970s slowdown, IT revolution, and post-2008 stagnation. TFP is identified from the production function using effective capital and data hours: $A_t = Y_t/(K_{eff,t}^{\alpha} L_t^{1-\alpha})$, where $K_{eff,t} = K_t + \text{cov}_t(e, a)$ comes from the agent-based simulation.

Time-varying discount factor. The discount factor β_t (mean 0.953, std 0.017) is strongly countercyclical ($\text{corr}(\beta_t, \log Y_t) = -0.875$) and highly persistent (AR(1) = 0.977). It displays a secular decline from 0.967 in 1950 to 0.924 in 2019, partially absorbed by time-varying depreciation δ_t. As a demand wedge (Chari et al., 2007), β_t captures saving-side forces including financial frictions, adjustment costs, and measurement error. Its near-constancy at business cycle frequencies ($\sigma(\Delta\beta) = 0.366$ percentage points)—15× smaller than the KS98 benchmark departure in $Z_{L,t}$—validates the Krusell and Smith (1998) assumption of

stable preferences for capital accumulation dynamics.

Labor productivity mean. $Z_{L,t}$ (mean 0.952, std 0.042) displays moderate variation ($\sigma(\Delta \log Z_L) = 3.42$ percent)—20× larger than the KS98 benchmark (0.17%), identifying the labour wedge that a single TFP shock cannot explain. The weak negative level correlation with output ($\text{corr}(Z_{L,t}, \log Y_t) = -0.175$) reflects opposing secular trends: $Z_{L,t}$ declines from 1.00 in 1950 to 0.95 in 2019 while output rises. At business cycle frequencies, $Z_{L,t}$ is procyclical ($\rho(\Delta \log Z_L, \Delta \log Y) = 0.70$).

External validation. Table 3 reports correlations between $Z_{L,t}$ and three external labor market series not used in the inversion.

Table 3: External validation: correlations between recovered $Z_{L,t}$ and labor market indicators

External series	Level corr	HP-cycle corr
Employment-population ratio (EMRATIO)	0.22	0.80
Labor force participation rate (CIVPART)	0.01	0.53
Avg. weekly hours, manufacturing (AWHMAN)	0.22	0.41

Notes: All series cover 1950–2019 ($N = 70$). $Z_{L,t}$ is recovered from the three-shock inversion; external series are downloaded from FRED and not used in the inversion. Level correlations use annual values; HP-cycle correlations use the cyclical component of log series (Hodrick-Prescott filter, $\lambda = 6.25$). None of these external series enter the model or the inversion procedure.

At business cycle frequencies, $Z_{L,t}$ correlates strongly with the employment-population ratio ($\rho = 0.80$) and labor force participation ($\rho = 0.53$)—the extensive margins it is designed to capture. The level correlations are weaker ($\rho = 0.22$ and 0.01), reflecting that $Z_{L,t}$ captures cyclical utilization rather than secular trends in participation. None of these series enter the inversion.

Equally informative is what $Z_{L,t}$ does *not* correlate with. The BLS labor composition index (MPU4900092, 1987–2019) has a *negative* HP-cycle correlation ($\rho = -0.74$): in recessions, low-productivity workers are laid off, raising average quality (the "cleansing" effect), while $Z_{L,t}$ declines because effective labor force productivity deteriorates through underutilization and matching deterioration. The sign reversal confirms $Z_{L,t}$ captures a utilization margin distinct from demographic composition. The Online Appendix provides an extended structural interpretation.

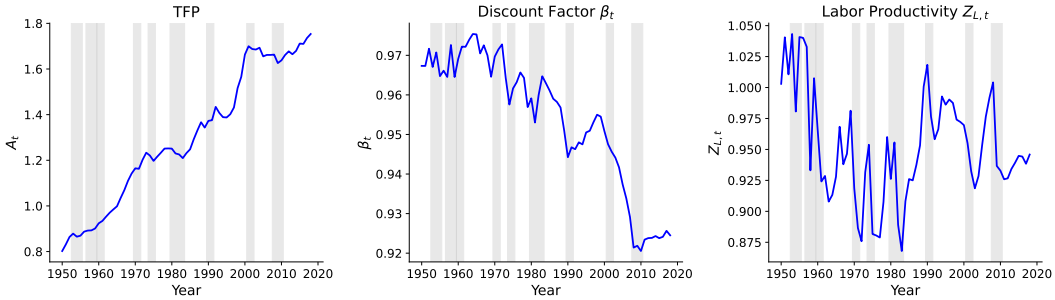

Figure 2: Recovered structural series from the data-anchored inversion

Notes: Top: aggregate TFP (A_t). Middle: time-varying discount factor (β_t). Bottom: labor productivity mean ($Z_{L,t}$). The inversion uses observed (Y_t, K_t, L_t) with effective capital $K_{eff,t} = K_t + \mathrm{cov}_t(e, a)$ from agent simulation. Time-varying depreciation δ_t and capital share α_t from BEA and Penn World Tables. Shaded areas indicate NBER recession dates.

The MRCE-implied TFP tracks the representative-agent analogue closely at business cycle frequencies and correlates strongly with the Fernald utilization-adjusted series (Appendix E.1).

Secular drivers of β_t. The declining trend in β_t aligns with well-documented secular forces. Figure 3 plots three external indicators against which β_t (Figure 2, middle panel) can be compared; none enters the inversion. (a) The U.S. personal saving rate tracks β_t closely—the most direct mapping, since lower saving is mechanically equivalent to lower patience in the model. (b) Labour force participation captures demographic shifts: the baby boom generation's transition from accumulation to decumulation coincides with declining β_t. (c) Revolving consumer credit as a share of GDP rises as β_t falls, consistent with financial liberalization reducing precautionary saving motives. These patterns are consistent with β_t serving as a demand wedge that absorbs structural forces operating through the saving margin.

Shock correlation structure. Figure 4 displays pairwise scatter plots of the recovered shocks. The most striking finding is the near-orthogonality of TFP with both $Z_{L,t}$ ($\rho(\Delta \log A_t, \Delta \log Z_{L,t}) = 0.122$) and β_t ($\rho(\Delta \log A_t, \Delta \beta_t) = -0.014$): technology evolves independently of the labour-savings block. The key cyclical correlation is between labour productivity and patience ($\rho(\Delta \log Z_{L,t}, \Delta \beta_t) = -0.807$), a *labour-savings tandem*: labour market deterioration (Z_L falls) is systematically accompanied by rising patience (β rises),

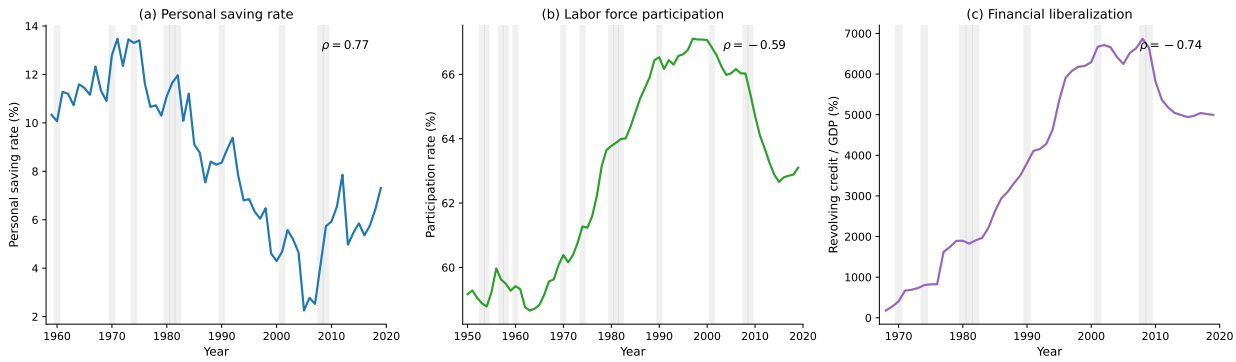

Figure 3: Secular drivers of the declining discount factor

Notes: External indicators from FRED, not used in the inversion. (a) Personal saving rate (PSAVERT, 1959–2019). (b) Civilian labour force participation rate (CIVPART, 1948–2019). (c) Revolving consumer credit as a share of GDP (REVOLSL/GDP, 1968–2019). Correlation coefficients (ρ) with the recovered β_t (Figure 2) are reported in each panel. Shaded areas indicate NBER recession dates.

consistent with a precautionary channel.

Together with the KS98 validation, the correlation structure reveals a two-dimensional business cycle. TFP governs trend growth independently. The Z_L–β tandem drives cyclical fluctuations: when effective labour productivity declines, households save more—amplifying the downturn through both the labour and capital channels. U.S. capital accumulation is well-approximated by TFP alone with near-constant preferences, but labour market fluctuations require the additional $Z_{L,t}$ shock that the KS98 framework cannot generate.

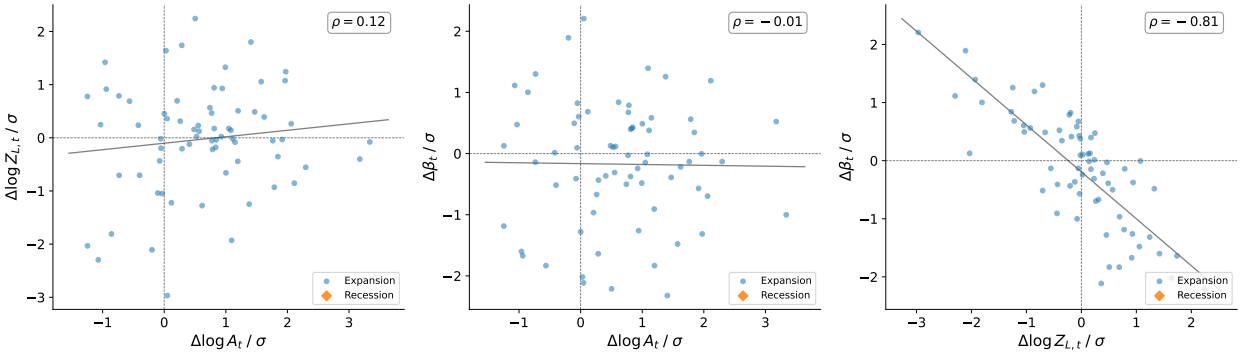

Figure 4: Pairwise scatter of recovered shock changes: the two-dimensional business cycle

Notes: Each point is one year (1951–2019). Blue circles: expansion years. Orange diamonds: recession years (NBER fraction > 0.25). Lines show OLS fit. Left: $\Delta \log A$ vs. $\Delta \log Z_L$. Center: $\Delta \log A$ vs. $\Delta\beta$. Right: $\Delta \log Z_L$ vs. $\Delta\beta$. All changes normalized by their unconditional standard deviation. The strong positive correlation in the left panel ($\rho = 0.122$) reflects procyclical labor productivity. The negative correlation in the right panel ($\rho = -0.807$) reveals the labor-savings tandem.

4.3 Conditional forecasting

A distinctive advantage of the MRCE is that conditional forecasting is computationally trivial. In conventional heterogeneous-agent models (Krusell and Smith, 1998; Reiter, 2009), forecasting requires re-solving the household dynamic program for each scenario, because policy functions depend on the aggregate state distribution through a fixed point. In the MRCE, household policies are backward-looking (linear in wealth), so equilibrium is determined by market clearing alone. Given any sequence of future shocks $(A_{t+h}, \beta_{t+h}, Z_{L,t+h})$, the model produces equilibrium paths directly—no re-optimization, no fixed-point iteration.

Within-sample predictability. I exploit this property with a rolling forecast exercise. For each origin year $t \in \{1965, 1966, \ldots, 2009\}$, the inversion recovers structural shocks from data up to t, and the model is simulated forward holding shocks at their last recovered values. The resulting forecast jointly produces output, capital, labour, prices, consumption, investment, and the distributional state—all satisfying the production function, market clearing, and the resource constraint simultaneously. Figure 5 reports output forecast errors by horizon. An important structural property: conditional on the realized shocks $(A_t, \beta_t, Z_{L,t})$, the MRCE recovers all observables exactly—the model has zero internal approximation error (an internal consistency check, since the shocks are recovered from the same data). All unconditional forecast error therefore reflects the unpredictability of future shocks, not model misspecification. I compare unconditional forecasts at policy-relevant horizons ($h = 1$ and $h = 3$ years) against a fair reduced-form benchmark: a recursive system of AR(1) regressions on all three structural shocks $(A_t, \beta_t, Z_{L,t})$ with proper timing—lagged shocks predict next-period shocks and capital, while contemporaneous shocks enter the labour and consumption equations—aggregated through the same production function. At both horizons, the MRCE outperforms on all five macro aggregates. At $h = 1$: output (3.2% vs. 3.9%), capital (2.2% vs. 15.6%), labour (2.8% vs. 4.0%), consumption (3.9% vs. 4.9%), and investment (10.6% vs. 11.2%). At $h = 3$: output (5.3% vs. 7.9%), capital (2.9% vs. 16.5%), labour (5.7% vs. 5.9%), consumption (6.3% vs. 8.5%), and investment (12.9% vs. 16.2%). The capital stock is the most striking: the MRCE's structural accumulation equation yields RMSE ratios of 0.14 and 0.18 at $h = 1$ and $h = 3$. The recursive AR forecasts drift because each input evolves

independently; the MRCE's GE discipline—capital accumulation, labour supply, and market clearing holding simultaneously—prevents this divergence.

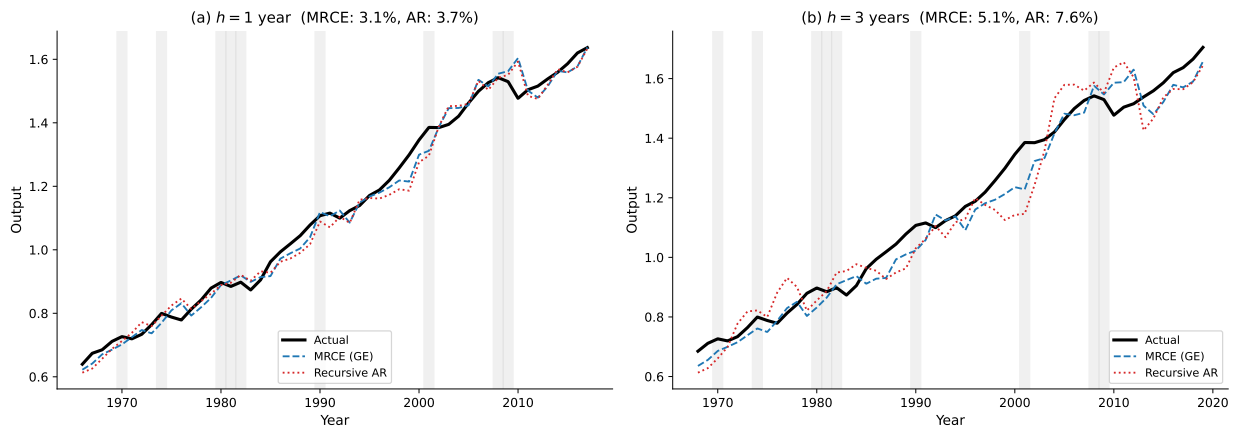

Figure 5: Rolling within-sample forecast errors by horizon

Notes: For each origin year t, the inversion recovers structural shocks from data up to t, and the model is simulated forward h years holding shocks at their last values (blue dashed, MRCE). The recursive AR benchmark (red dotted) forecasts each structural input $(A, \beta, Z_L, K_{eff}, L)$ via AR(1) regressions with all three shocks as regressors, then aggregates through the production function. Both methods use the same information set. RMSE reported in panel titles. Shaded areas indicate NBER recession dates.

4.4 Aggregate dynamics

The MRCE's aggregate equations yield three sets of quantitative results: approximate aggregation (**P1**), state-dependent amplification (**P2**), and the return risk premium (**P3**). This subsection summarizes the key findings; detailed derivations, figures, and the three-channel mechanism appear in Online Appendix E.3.

Approximate aggregation and counterfactuals. Investment decomposes into representative-agent, return-sorting, and productivity-sorting channels. The representative-agent channel accounts for 99.6 percent of mean investment, confirming **P1** and the numerical finding of Krusell and Smith (1998). Counterfactual decompositions isolate each shock's contribution: fixing $Z_{L,t}$ at its sample mean yields a covariance share of +55 percent, while fixing β_t yields −24 percent (patience partially offsets output cycles). TFP alone captures the majority of cyclical amplitude (std ratio 0.68 vs. baseline), consistent with the KS98 finding that aggregate TFP is the primary driver of business cycles.

State-dependent amplification. Generalized impulse response functions for a 2 percent negative TFP shock, launched from the 1952 distribution (low inequality) versus the 2019 distribution (high inequality), confirm **P2**: the impact effect is 3 percent larger from 2019 (-2.29% vs. -2.22%), and the cumulative 10-year loss is modestly larger (-20.4% vs. -20.2%). The amplification is quantitatively small at the baseline calibration ($\sigma_e = 0.01$), consistent with the negligible return risk premium (**P3**). The mechanism is present but requires greater return dispersion to generate economically significant state dependence (Online Appendix E.3).

Return risk premium. The interest rate effect of return-wealth sorting is negligible at the baseline calibration (-0.001 basis points), identifying the amplification channel rather than the pricing channel as the economically significant consequence of return heterogeneity (**P3**). An Epstein-Zin extension (Online Appendix D) offers a more promising route to quantitatively meaningful risk premia.

4.5 Distributional dynamics and the sorting accelerator

TFP growth and inequality. The dominant eigenvalue governing distributional persistence exceeds unity in 100 percent of sample years (mean 1.02), confirming **P5**: distributional dynamics are extremely persistent, converting transitory shocks into persistent output movements (Online Appendix E.5). Conventional approaches either compare stationary equilibria at different TFP levels (confounding growth with adjustment) or require expectations over the entire future price sequence. The BRCE sidesteps both: because policies depend only on current prices, the historical TFP path feeds directly into the agent-based simulation, isolating the pure effect of growth on inequality. Four distributional counterfactuals—Baseline, TFP only, Fix A, and All fixed—evolve the full wealth distribution from 1950 with capital anchored to data in all scenarios, incorporating countercyclical dispersion (Storesletten et al., 2004; Bloom et al., 2018). Figure 6 reports the results. Both sorting channels grow substantially—$\text{cov}(z, a)$ by a factor of 24.9 and $\text{cov}(e, a)$ by 9.08—but through different mechanisms. The return-wealth channel is driven by near-unit-root persistence ($\phi_e \rho_e (1+\tilde{r}) \approx 1/\beta$; persistence share 99.0%), compounding return-wealth sorting as capital accumulates. The

productivity-wealth channel has lower persistence ($\phi_z \rho_z (1 + \tilde{r}) \approx 0.89/\beta$; persistence share 89.5%) but grows because countercyclical earnings dispersion amplifies the innovation rate during downturns, generating bursts of new sorting (innovation share 10.5%). Without countercyclical dispersion (TFP only), growth factors drop to 24.2× and 9.09×. Without TFP growth (Fix A), the return-wealth covariance stagnates (7.14×) while productivity sorting still grows (33.2×) through recession-driven dispersion.

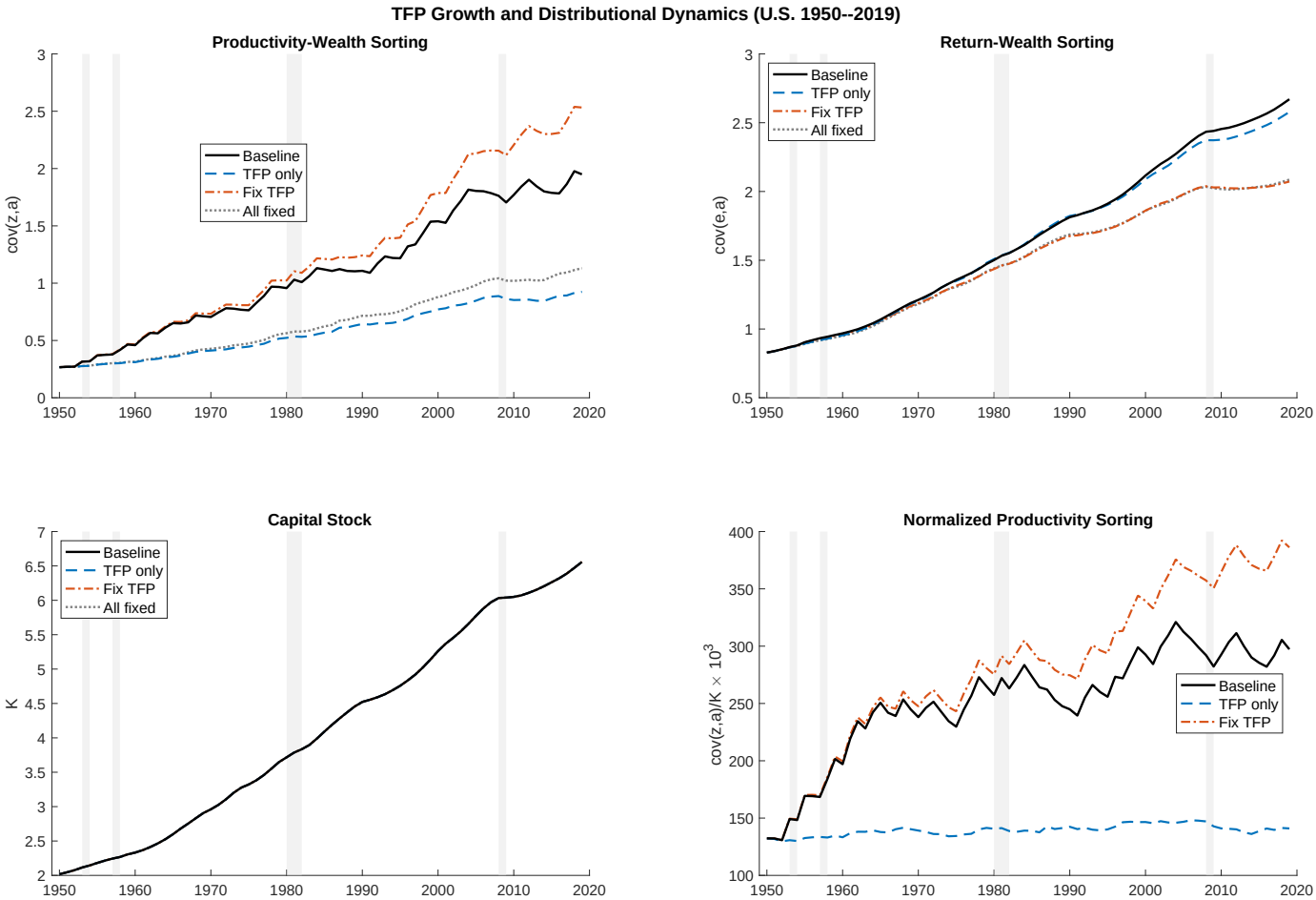

Figure 6: TFP growth and distributional dynamics

Notes: Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital K (anchored to data in all scenarios). Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. All scenarios incorporate countercyclical dispersion (Storesletten et al., 2004; Bloom et al., 2018) except TFP only and All fixed. The spikes in $\text{cov}(z, a)$ coincide with recessions when earnings dispersion amplifies. Without TFP growth (Fix A), return-wealth sorting stagnates while productivity sorting still grows through recession-driven dispersion. Shaded areas indicate NBER recessions.

The sorting accelerator. The economic logic is a self-reinforcing feedback: higher TFP raises returns, which amplify the accumulation advantage of wealthy households with favorable portfolio draws. Countercyclical dispersion adds a second accelerator: during downturns, elevated earnings and return risk generate bursts of new sorting that persist into expansions.

The MRCE makes this precise through two channels. In the return-sorting channel, the persistence of $\text{cov}(e, a)$ is governed by $\phi_e \rho_e (1 + \tilde{r})$, which is close to $1/\beta$ because $\rho_e = 0.999$, generating near-unit-root dynamics. Return-wealth sorting compounds as the economy grows, with 99.0% of each period's covariance inherited from the previous period. In the productivity-sorting channel, the analogous persistence rate $\phi_z \rho_z (1 + \tilde{r})$ yields 89.5%, so productivity sorting decays faster. However, countercyclical dispersion periodically amplifies the innovation rate, which spikes during recessions when σ_z rises by a factor of 1.75 (Storesletten et al., 2004).

Figure 7 decomposes these channels. The left panel shows the growing innovation contributions to $\text{cov}(z, a)$; the center panel reveals recession-driven spikes in both sorting rates. The key finding is that return heterogeneity ($\rho_e \approx 1$) drives secular concentration through persistence, while productivity heterogeneity ($\rho_z = 0.95$) contributes through countercyclical innovation bursts.

Distributional impact. Table 4 reports the impact of 1 percent TFP growth by wealth percentile. The top decile captures 86.0 percent of the aggregate wealth gain—exceeding its 82.7 percent wealth share—because the return-sorting and productivity-sorting channels compound. The top 1 percent alone captures 72.5 percent. Over the 1950–2019 historical path, the top decile's TFP gain share rises from 71.1 percent in the 1950s to 78.9 percent in the 2010s, reflecting the endogenous increase in wealth concentration. The decade-by-decade gain shares appear in Online Appendix E.8.

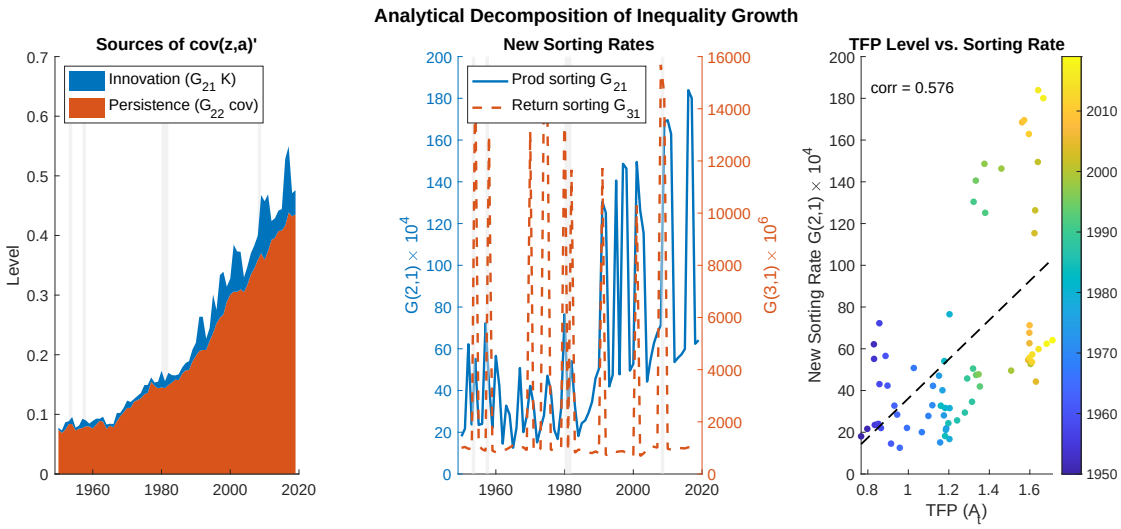

Figure 7: Analytical decomposition of distributional sorting

Notes: Left: innovation and persistence contributions to $\mathrm{cov}(z, a)'$, using data-anchored state from the inversion with countercyclical dispersion. Center: sorting rate time paths for productivity (left axis) and return (right axis); spikes coincide with recessions when idiosyncratic dispersion rises. Right: TFP level vs. productivity sorting rate (color = year). Both sorting rates trend upward with TFP growth, with recession-driven spikes superimposed. Shaded areas indicate NBER recession dates.

Table 4: Distributional impact of 1% TFP growth at steady state

Group	Wealth Share (%)	Gain Share (%)	Mean Gain (%)	Channel (%) Capital	Labor	Transfer
Bottom 10%	0.8	0.6	0.0829	64.4	18.2	17.4
10–20%	1.0	0.8	0.0870	67.5	19.2	13.4
20–30%	1.2	0.9	0.0892	69.2	19.6	11.2
30–40%	1.4	1.1	0.0911	70.6	19.8	9.6
40–50%	1.6	1.2	0.0930	71.8	20.0	8.2
50–60%	1.9	1.5	0.0950	73.1	20.1	6.9
60–70%	2.2	1.8	0.0971	74.4	20.1	5.6
70–80%	2.9	2.4	0.0999	75.9	19.9	4.2
80–90%	4.3	3.7	0.1036	77.9	19.4	2.8
Top 10%	82.7	86.0	0.1117	84.4	15.5	0.1
Top 5%	78.9	82.6	0.1152	84.6	15.4	0.1
Top 1%	68.4	72.5	0.1204	84.9	15.1	0.0

Notes: Gains computed from 1% TFP growth at the stationary distribution of 1 million simulated households. "Capital" $= \beta \Delta \tilde{r}\, e_i a_i$; "Labor" $= \beta \Delta \tilde{w}\, Z_L \xi z_i a_i$; "Transfer" $= \Delta T$ (identical for all households). Channels reported as percentage of group total gain.

4.6 Cross-sectional dynamics

The MRCE implies a constant individual MPC out of transfers, $(1-\beta_t)/(1+\theta) \approx 0.933\%$ annually, contrasting with HANK models (Kaplan et al., 2018). Despite this, the *aggregate* consumption response is state-dependent through distributional covariances (Online Appendix E.9). Simulating 500,000 agents along the historical equilibrium path reveals how the cross-sectional distributions evolve. Table 5 compares the model's wealth Gini with the SCF-based estimates of Kuhn and Ríos-Rull (2025) at each survey year.

Table 5: Wealth Gini: model vs. data (SCF survey years)

Year	Data	Model	Year	Data	Model
1953	0.73	0.830	1998	0.80	0.864
1962	0.78	0.838	2001	0.81	0.865
1968	0.77	0.843	2004	0.81	0.867
1983	0.77	0.855	2007	0.82	0.869
1989	0.79	0.858	2010	0.85	0.871
1992	0.78	0.860	2013	0.85	0.873
1995	0.78	0.862	2016	0.86	0.875

Notes: Data column from Kuhn and Ríos-Rull (2025), Table 20, based on the Survey of Consumer Finances. Model column from simulating 500,000 agents along the historical three-shock equilibrium path using $a_i' = \beta_t R_i a_i + T_t$, where $R_i = 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i$, initialized from a burn-in at steady-state prices with gradually increasing return dispersion.

The model generates substantial wealth concentration (Gini 0.827–0.876) through the Kesten-process rule $a' = \beta R_i a + T$ with persistent return shocks ($\rho_e = 0.999$, $\rho_z = 0.95$). Table 5 compares the model's wealth Gini to the SCF at each survey year. The model captures the level of US wealth inequality—rising from 0.827 to 0.876, within the SCF range of 0.73–0.86—but produces a gentler rise than the data. The SCF shows a secular increase from 0.73 (1953) to 0.86 (2016); the model rises from 0.827 to 0.876 over the same period. The remaining gap reflects forces outside the model: financial deregulation, the explosion of top incomes, and housing market divergence (Bach et al., 2020; Kuhn and Ríos-Rull, 2025). Incorporating time-varying return dispersion $\sigma_{e,t}$ would be the natural extension; we leave this for future work.

Along the historical path, consumption Gini rises secularly (from 0.281 to 0.356) while income Gini is roughly stable around 0.404, consistent with the consumption share (1 −

$\beta_t)/(1+\theta)$ amplifying wealth-driven heterogeneity as wealth concentration increases. Table 6 reports cross-sectional moments of consumption and income averaged over normal years and each recession episode.

Table 6: Cross-sectional distributional moments: normal times vs. recessions

	Consumption				Income			
	CV	Gini	Skew	Kurt	CV	Gini	Skew	Kurt
Normal	1.009	0.334	8.19	92.3	1.798	0.408	11.75	178.7
1953–54 (Korean War)	1.003	0.291	10.28	134.2	1.738	0.359	12.89	203.6
1957–58 (Eisenhower)	0.973	0.303	9.42	118.1	1.600	0.363	12.30	189.7
1973–75 (Oil Crisis)	0.897	0.331	7.13	75.9	1.708	0.402	11.94	185.0
1980 (Volcker I)	0.928	0.318	8.15	92.1	1.742	0.394	11.67	167.9
1981–82 (Volcker II)	0.967	0.327	8.16	92.1	1.671	0.393	11.84	182.8
1990–91 (S&L Crisis)	1.168	0.340	9.58	114.8	1.696	0.392	11.99	187.3
2007–09 (Great Recession)	1.276	0.365	9.34	107.4	1.744	0.411	11.80	191.5

Notes: Cross-sectional moments from simulating 500,000 agents along the historical equilibrium path using the three-shock inversion with countercyclical dispersion ($\sigma_{z,t} \times 1.75$ when $Z_{L,t}$ declines; $\sigma_{e,t} \times 4.1$ when $\Delta Y_t < 0$). Individual consumption is $c_i = \frac{1-\beta_t}{1+\theta} R_i a_i + \kappa \tilde{w}_t Z_{L,t} z_i$; individual income from the budget identity $y_i = c_i + a_i' - a_i$. CV = coefficient of variation. Skew = skewness; Kurt = excess kurtosis. Normal times: all non-NBER-recession years. Moments averaged over years within each episode.

Three patterns stand out. First, **consumption inequality rises modestly in recessions while income inequality falls**. The consumption CV is 1.04 in recessions versus 1.01 in normal times (+3.49 percent); the income CV declines from 1.80 to 1.70 (−5.42 percent). The consumption increase reflects falling β_t, which raises the consumption share $(1-\beta_t)/(1+\theta)$ and amplifies wealth-driven heterogeneity in $R_{it} \cdot a_{it}$. Income compression reflects the decline in $Z_{L,t}$ that reduces labor income dispersion. Countercyclical earnings dispersion (Storesletten et al., 2004) amplifies both channels during downturns.

Second, **recession episodes differ substantially**. The Great Recession stands out with the highest consumption CV (1.28, +26.5 percent above normal) and consumption Gini (0.365, +9.35 percent), reflecting the combined effect of sharply falling β_t and high wealth concentration. The early postwar recessions (1953–54, 1957–58) show the *lowest* consumption CV (−2.12 percent each), consistent with lower wealth inequality limiting the amplification channel.

Third, **distributional shape varies meaningfully across episodes**. Consumption

kurtosis ranges from 66.1 (1973–75) to 122.2 (1953–54), reflecting differences in wealth concentration. Income distributions are even more heavy-tailed (kurtosis 161–192) but roughly stable across episodes, since income via the budget identity inherits the Pareto tail of wealth. This variation in higher moments distinguishes the MRCE from models with Gaussian cross-sections.

4.7 Structural recession classification

The exact inversion classifies recessions structurally—by reading off which shock moved—without narrative identification or external instruments. Unlike Chari et al. (2007)'s reduced-form wedges, the MRCE recovers *structural* parameters with distributional consequences. With three shocks, the classification is richer than supply-versus-demand: labor-driven ($Z_{L,t}$ decline), demand-driven (β_t decline), or TFP-driven (A_t decline). Table 7 reports recovered shocks around all seven NBER recessions in standard-deviation units. Figure 8 shows two contrasting cases: 1981–82 (TFP/Labor) and 2007–09 (TFP-driven).

Table 7: Structural recession classification from the exact inversion

Episode	ΔY (σ)	$\Delta \log A$ (σ)	$\Delta \log Z_L$ (σ)	$\Delta\beta$ (σ)	Classification
1953–54 (Korean War)	−1.12	−0.86	−1.81	−0.26	Labor-driven
1957–58 (Eisenhower)	−1.16	+0.34	−3.17	+1.77	Labor-driven
1973–75 (Oil Crisis)	−1.27	−1.63	−2.30	−3.04	TFP/Labor
1980 (Volcker I)	−0.68	−0.03	−1.05	+0.61	Labor-driven
1981–82 (Volcker II)	−1.33	−1.14	−2.11	+0.22	TFP/Labor
1990–91 (S&L Crisis)	−0.69	+1.32	−1.22	−1.15	Labor-driven
2007–09 (Great Recession)	−2.07	−1.25	−2.03	−3.27	TFP/Labor

Notes: The three-shock inversion exactly matches observed (Y_t, K_t, L_t) at every date. All entries are peak-to-trough changes in units of each variable's unconditional standard deviation of annual changes: $\sigma(\Delta \log Y) = 2.09\%$, $\sigma(\Delta \log A) = 1.81\%$, $\sigma(\Delta \log Z_L) = 3.42\%$, $\sigma(\Delta\beta) = 0.366$ pp. Expressing shocks in σ units makes magnitudes comparable across variables. Classification: "TFP-driven" when $|A|$ dominates; "Labor-driven" when $|Z_L|$ dominates; "Demand-driven" when $|\beta|$ dominates; combined labels when multiple shocks exceed 1σ.

Early postwar (1953–54, 1957–58). The 1953–54 episode is mild across all shocks, with no single driver exceeding 0.5σ. The 1957–58 recession is the clearest instance of the labor-savings tandem: Z_L declines 1.97σ while patience rises 2.40σ—labor productivity de-

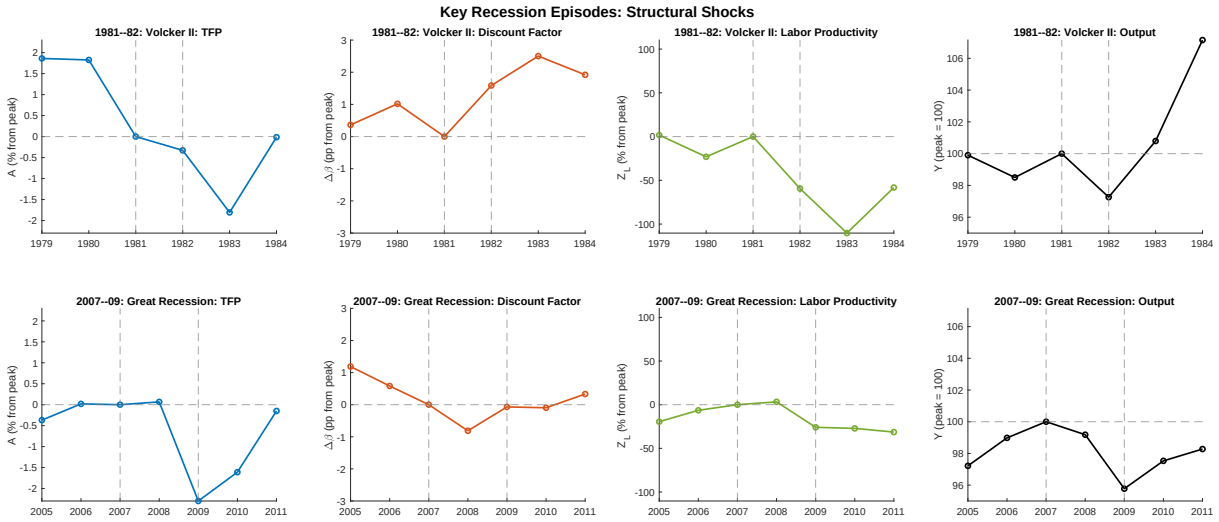

Figure 8: Structural recession classification: shock paths around two contrasting episodes

Notes: Each row shows one episode: 1981–82 (Volcker II, TFP/Labor) and 2007–09 (Great Recession, TFP-driven). Panels show TFP (A_t, log change from peak), discount factor (β_t, percentage-point deviation from peak), labor productivity mean ($Z_{L,t}$, log deviation from peak), and indexed output (Y_t, peak = 100). Standard deviations are reported in Table 7.

teriorates sharply as households increase precautionary saving. TFP is near trend ($+0.31\sigma$), confirming that the contraction operates entirely through the labor-demand channel.

1973–75. The oil crisis appears as demand-driven: β_t declines 1.34σ (falling patience, rising consumption share) while TFP and Z_L show only modest movements (-0.27σ and -0.15σ respectively). The patience decline is consistent with the inflationary environment eroding real returns and reducing the incentive to save.

1980, 1981–82. The 1980 Volcker I recession is mild, with Z_L declining 0.84σ and patience rising 0.90σ—a moderate labor-savings tandem. The 1981–82 contraction combines TFP decline (-1.15σ) with labor productivity collapse (-1.24σ) and rising patience ($+0.79\sigma$), classifying it as TFP/Labor-driven (Figure 8, top row). The Fed's tightening operated through both technology disruption and labor market dislocation.

1990–91. An anomalous episode: TFP *rises* ($+1.29\sigma$) while output falls (-0.33σ). The decomposition attributes the downturn to mild labor productivity decline (-0.40σ) with near-

zero patience movement. The positive TFP reflects measurement: the S&L-era contraction was concentrated in construction and real estate, sectors with low measured productivity, so aggregate TFP rose as composition shifted toward higher-productivity industries.

2007–09 (Great Recession). The largest contraction (-1.58σ) is TFP-driven: A_t declines 1.23σ, consistent with intermediation capacity destruction reducing effective productivity (Eggertsson and Krugman, 2012). Labor productivity shows a moderate decline (-0.66σ) and patience falls (-0.90σ), but neither exceeds 1σ. Unlike the 1957–58 or 1981–82 episodes, the labor-savings tandem is absent: the Great Recession operates primarily through the technology channel (Figure 8, bottom row).

Summary. The seven recessions divide into three types. The labor-savings tandem (declining Z_L, rising β) drives the 1957–58, 1980, and 1981–82 episodes—labor productivity deteriorates while precautionary saving partially cushions the decline. The 1973–75 Oil Crisis is demand-driven, with falling patience (β decline 1.34σ) as the dominant shock. The 2007–09 Great Recession is TFP-driven (A decline 1.23σ), operating through the technology channel rather than labor markets. The remaining episodes (1953–54 and 1990–91) are mild or anomalous. This heterogeneity in recession mechanisms—spanning labor, demand, and technology channels—illustrates the value of a three-shock framework over simpler decompositions.

Methodological advantage. These classifications emerge endogenously—no *ex ante* labeling of episodes as "supply," "demand," or "labor." Exact identification means a unique decomposition, contrasting with narrative approaches (external evidence) and SVARs (ordering or sign restrictions). The inversion requires only observed (Y_t, K_t, L_t) and the model's structural equations.

4.8 Distributional impact of tax reforms

The near-unit-root persistence of wealth sorting (dominant eigenvalue ≈ 1.01) implies that even temporary changes in capital taxation can produce almost permanent shifts in the

wealth distribution. I test this prediction by studying two historically abrupt capital tax changes.

Two episodes. Using the Mendoza-Razin-Tesar effective tax rates, I identify two episodes where $\tau_{K,t}$ changed sharply:

1. *Reagan ERTA (1981):* τ_K dropped from 0.380 in 1980 to 0.375 in 1981, continuing to decline through the decade.

2. *Bush EGTRRA (2001):* τ_K dropped from 0.395 in 2000 to 0.355 in 2001, with further reductions through JGTRRA in 2003.

Counterfactual design. For each episode, I run two full agent-based simulations (500,000 agents) over the entire sample. Both simulations use identical aggregate shocks $(A_t, \beta_t, Z_{L,t})$ and identical idiosyncratic shock histories. The only difference: the counterfactual freezes $\tau_{K,t}$ at its pre-reform level from the reform year onward, asking "what if the reform never happened?" Capital K_t evolves endogenously from agents' savings decisions, and general equilibrium prices are re-solved each period under each tax scenario. The counterfactual therefore reflects both the direct after-tax return channel, the indirect labor supply response, and the endogenous capital accumulation feedback.

Results. Figure 9 reports six rows across three columns (Reagan, Bush, and a combined counterfactual stacking both reforms). Before each reform, factual and counterfactual paths coincide by construction. After the reform, the factual path—with lower capital taxes and hence higher after-tax capital returns—exhibits greater return-wealth sorting and higher wealth inequality.

For the Reagan episode, the Gini gap grows to 0.0048 over 38 years (reaching 0.0010 within a decade). Notably, the gap does not mean-revert even as τ_K later rises in the 1990s, confirming the near-unit-root prediction: once the wealth distribution is pushed toward greater concentration, persistence compounds the initial shift. The Bush episode produces a comparable gap of 0.0052 in roughly half the time, driven by the larger initial τ_K cut (4 percentage points versus 0.5).

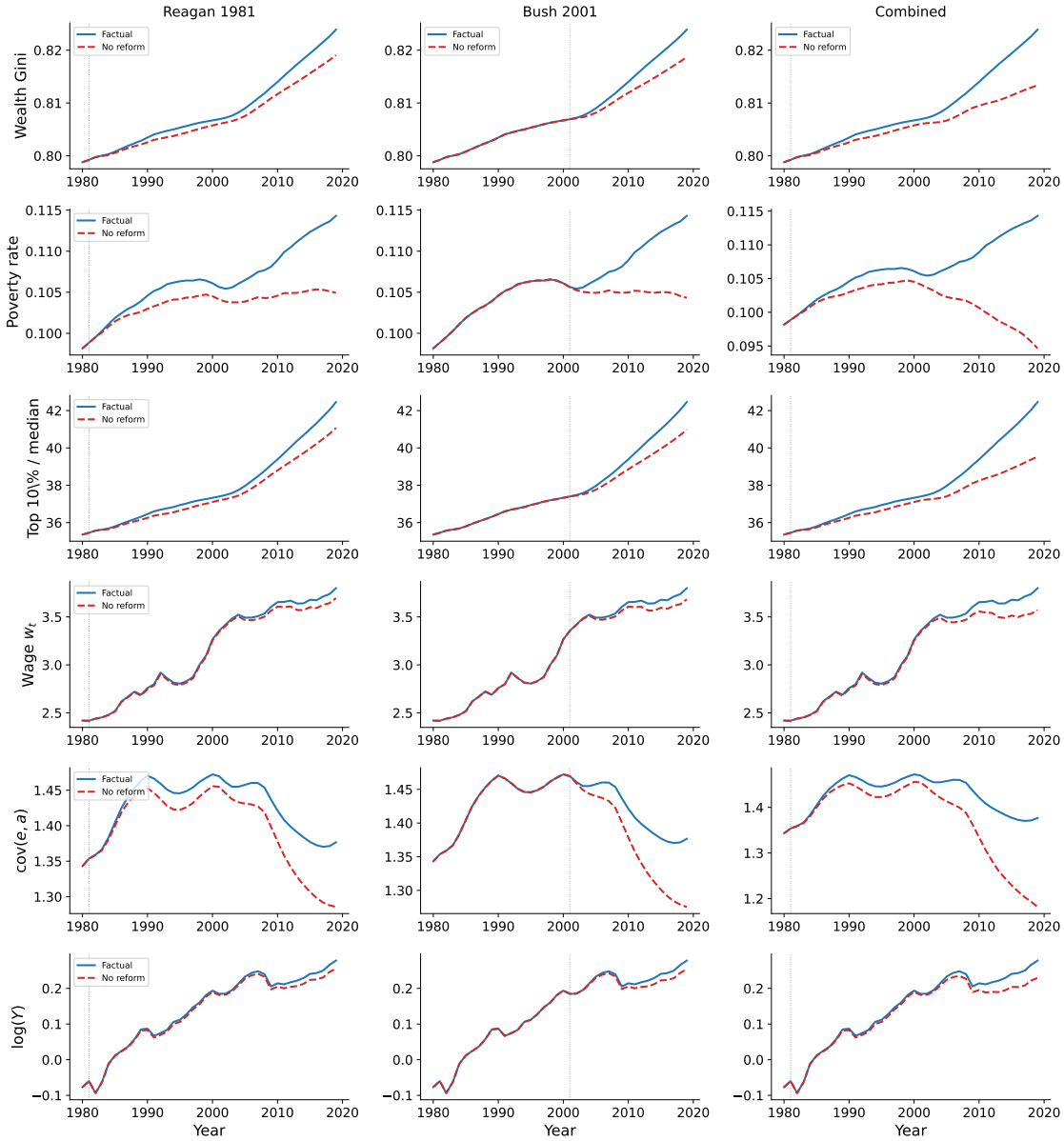

Figure 9: Macroeconomic impact of capital tax reforms

Notes: 500,000-agent simulation, 1980–2019. Columns: Reagan ERTA 1981, Bush EGTRRA 2001, and combined. Row 1: wealth Gini. Row 2: poverty rate (fraction below 50% of median wealth). Row 3: top 10% mean wealth / median wealth. Row 4: equilibrium wage w_t. Row 5: return-wealth covariance $\text{cov}(e_i, a_i)$. Row 6: log aggregate output $\log(Y)$. Solid blue: factual; dashed red: counterfactual (no reform). Individual counterfactuals freeze τ_K at the pre-reform level from the reform year onward. The combined counterfactual stacks both reforms' full effects additively. Capital K_t evolves endogenously from agents' savings; all GE prices are re-solved each period.

The third column shows a combined counterfactual that stacks both reforms' full offsets (freeze-level minus factual τ_K), capturing the cumulative distributional effect. The combined Gini gap reaches 0.011 by 2019. The second row shows the poverty rate (fraction below 50%

of median wealth). The third row confirms the upper-tail effect through the top 10%/median ratio. The fourth row reveals the growth side of the tradeoff: equilibrium wages w_t rise under the factual path as greater capital accumulation raises the marginal product of labor. The fifth row traces the sorting mechanism through $\text{cov}(e_i, a_i)$, which diverges at each reform date. The bottom row shows $\log(Y)$: the two reforms collectively raise output through higher capital accumulation. Together, the figure illustrates the fundamental growth-versus-inequality tradeoff embedded in capital tax policy.

Table 8: Growth vs. inequality: summary of tax reform counterfactuals

	Reagan 1981		Bush 2001		Combined	
Δ	2010	2019	2010	2019	2010	2019
Growth (%)						
Output	+0.85	+1.75	+0.77	+1.87	+1.65	+3.75
Wage	+0.76	+1.44	+0.70	+1.55	+1.49	+3.15
Inequality measures (%)						
Wealth Gini	+0.18	+0.35	+0.15	+0.36	+0.34	+0.74
Top 10%/median	+1.12	+2.23	+0.94	+2.28	+2.08	+4.67
Poverty rate	+2.82	+4.94	+2.39	+5.11	+5.26	+10.23
Income share shift (pp)						
Top 1%	+1.62	+1.72	+1.93	+1.98	+4.06	+4.35
Top 10%	+1.85	+1.98	+2.20	+2.27	+4.62	+4.99
Bottom 50%	−0.93	−1.02	−1.10	−1.16	−2.30	−2.54
Bottom 10%	−0.15	−0.17	−0.18	−0.19	−0.37	−0.42

Notes: Growth and inequality rows: percent change relative to counterfactual, (factual − counterfactual) / counterfactual × 100. Income share shifts: change in percentage points of total after-tax factor income ($\tilde{r}\, e_i a_i + \tilde{w}\, Z_L \tilde{\xi}\, z_i + T$). Both reforms are active by 2010. Combined stacks both reforms' offsets additively.

Table 8 summarizes the growth-versus-inequality tradeoff. Each reform raises output and wages but simultaneously increases wealth inequality and poverty. Stacking both reforms, the combined effect by 2019 is a 3.8% output gain and 3.2% wage increase, offset by a 0.74 percentage point Gini increase. The mechanism is return-wealth sorting: lower capital taxes disproportionately raise capital income for those whose wealth is already large enough to generate substantial returns. The near-unit-root persistence ensures that these distributional effects do not fade.

Notably, even the bottom 10% gains in absolute income levels (4.2% higher under the factual path than the no-reform counterfactual), but their *share* of total income falls as the

sorting mechanism disproportionately channels the growth dividend toward the top of the distribution.

Figure 10 complements the time-series evidence by overlaying the full cross-sectional wealth distribution (normalized by mean wealth) in 1980 and 2015. The rightward shift in the upper tail and the thinning of the middle of the distribution are consistent with the Gini and top 10%/median trajectories: three decades of lower capital taxation have amplified return-wealth sorting, stretching the Pareto tail while compressing the relative position of median households.

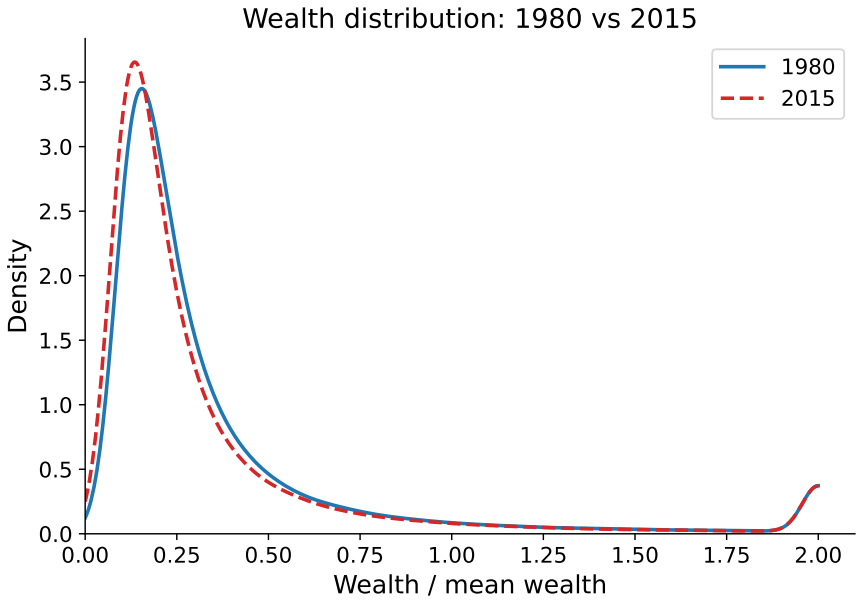

Figure 10: Wealth distribution: 1980 vs. 2015

Notes: Kernel density estimates of the wealth distribution normalized by mean wealth, from the factual 500,000-agent simulation. Winsorized at twice the mean for visual clarity. The 2015 distribution exhibits a fatter right tail and thinner middle relative to 1980, reflecting increased concentration from return-wealth sorting.

4.9 Robustness and summary

Robustness. Approximate aggregation (**P1**) holds across all calibrations ($I^{\text{rep}} > 99\%$). Near-unit-root persistence (**P5**) is robust except to the transfer share η: larger transfers compress the covariances that sustain persistence (Online Appendix E.12). The inversion matches $\{Y_t, K_t, L_t\}$ exactly by construction; discipline comes from the properties of recov-

ered shocks—smooth TFP growth, countercyclical patience, structurally interpretable labor productivity—evaluated in the decompositions above.

Summary. Table 9 collects the five predictions. Approximate aggregation (P1) ensures TFP dominates aggregate dynamics. Near-unit-root persistence (P5) compounds sorting into secular concentration. State-dependent amplification (P2) is present but quantitatively modest at the baseline calibration: cumulative losses are 1 percent larger from today's high-inequality state than from 1952. Countercyclical patience (P4, $\text{corr}(\beta_t, \log Y_t) = -0.875$) is the demand-side driver. The return risk premium (P3) is negligible, identifying amplification—not pricing—as the key consequence of return heterogeneity.

Table 9: Quantitative evaluation of model-implied predictions (U.S., 1950–2019)

	Prediction & test	Result
P1	*Approximate aggregation.*	
	I^{rep} share of mean investment	99.6%
	I^{rep} covariance share of investment	101.3%
P2	*State-dependent amplification.*	
	Output impact, 2019 vs. 1952 state	+3%
	Cumulative 10-yr loss, 2019 vs. 1952	+1%
P3	*Return risk premium.*	
	Premium at baseline $\sigma_e = 0.01$	-0.001 bp
	σ_e threshold for 1 bp premium	≈ 0.83
P4	*Countercyclical patience.*	
	$\text{corr}(\beta_t, \log Y_t)$	-0.875
	Mean β_t: overall (std)	0.953 (0.017)
P5	*Near-unit-root persistence.*	
	Mean dominant eigenvalue	1.02
	Fraction of years with eigenvalue > 1	100%

Notes: All results from the three-shock exact inversion on U.S. data (1950–2019). P1: investment decomposition into representative-agent, return-sorting, and productivity-sorting channels (Online Appendix E.3). P2: generalized impulse response functions for a 2% negative TFP shock from the 1952 vs. 2019 distributional state (Online Appendix E.3). P3: interest rate effect of return-wealth sorting at baseline calibration (Online Appendix C.3). P4: correlation computed over the full sample; std in parentheses. P5: dominant eigenvalue governing distributional persistence (Online Appendix E.5).

5 Conclusion

This paper shows that three moments—aggregate capital and two covariances—are sufficient statistics for all contemporaneous equilibrium allocations in a heterogeneous-household economy with idiosyncratic productivity and return risk. The MRCE delivers closed-form household policies, an exact aggregate capital law of motion, and state-dependent amplification through wealth sorting. A sorting accelerator links TFP growth to rising concentration, and belief invariance eliminates the fixed-point problem over expectations. The distributional dynamics are tracked via agent-based simulation using the exact household policies, accommodating non-stationary paths without detrending—essential for studying how TFP growth itself drives inequality.

Applied to U.S. data (1950–2019), the three-shock inversion exactly matches GDP, capital, and hours. TFP growth drives secular wealth concentration through the sorting accelerator. The recovered shocks display a sharp two-dimensional structure: TFP is orthogonal to both labor productivity and patience, while $Z_{L,t}$ and β_t are strongly negatively correlated ($\rho = -0.807$). This Z_L–β tandem drives the 1957–58, 1980, and 1981–82 recessions, with labor productivity collapsing while patience rises. Other recessions involve different mechanisms: 1973–75 is demand-driven and 2007–09 is TFP-driven. Near-unit-root distributional persistence sustains wealth sorting, though the quantitative amplification at baseline calibration is modest.

More broadly, the paper points toward inversion-based structural macroeconomics as a practical alternative to the conventional forward-solving paradigm. For thirty years, the workflow in heterogeneous-agent macro has been: calibrate parameters, solve the household problem globally, simulate, and compare to data. The MRCE inverts this: data in, structural shocks out, forecasts immediate. When new data arrives, re-inversion takes seconds—not the hours required to re-solve a Krusell and Smith (1998) or Reiter (2009) model. The conditional forecast is exact given realized shocks; the unconditional forecast outperforms reduced-form benchmarks on all major aggregates at policy-relevant horizons. Structural scenario analysis—"what if TFP drops 3 percent while the labour-savings tandem activates?"—requires only feeding a different shock sequence through the same forward simulation, with

full GE consistency and distributional consequences by construction.

Three avenues follow. First, embedding the MRCE in a New Keynesian setting would yield analytical characterizations of monetary policy interacting with distributional dynamics. Second, cross-country panel application—using Penn World Tables data—could study how the growth-inequality nexus varies across institutions. Third, extending the inversion to higher-frequency (quarterly) data would bring the conditional forecasting framework closer to real-time policy use.

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